

Homework 5

Instructions

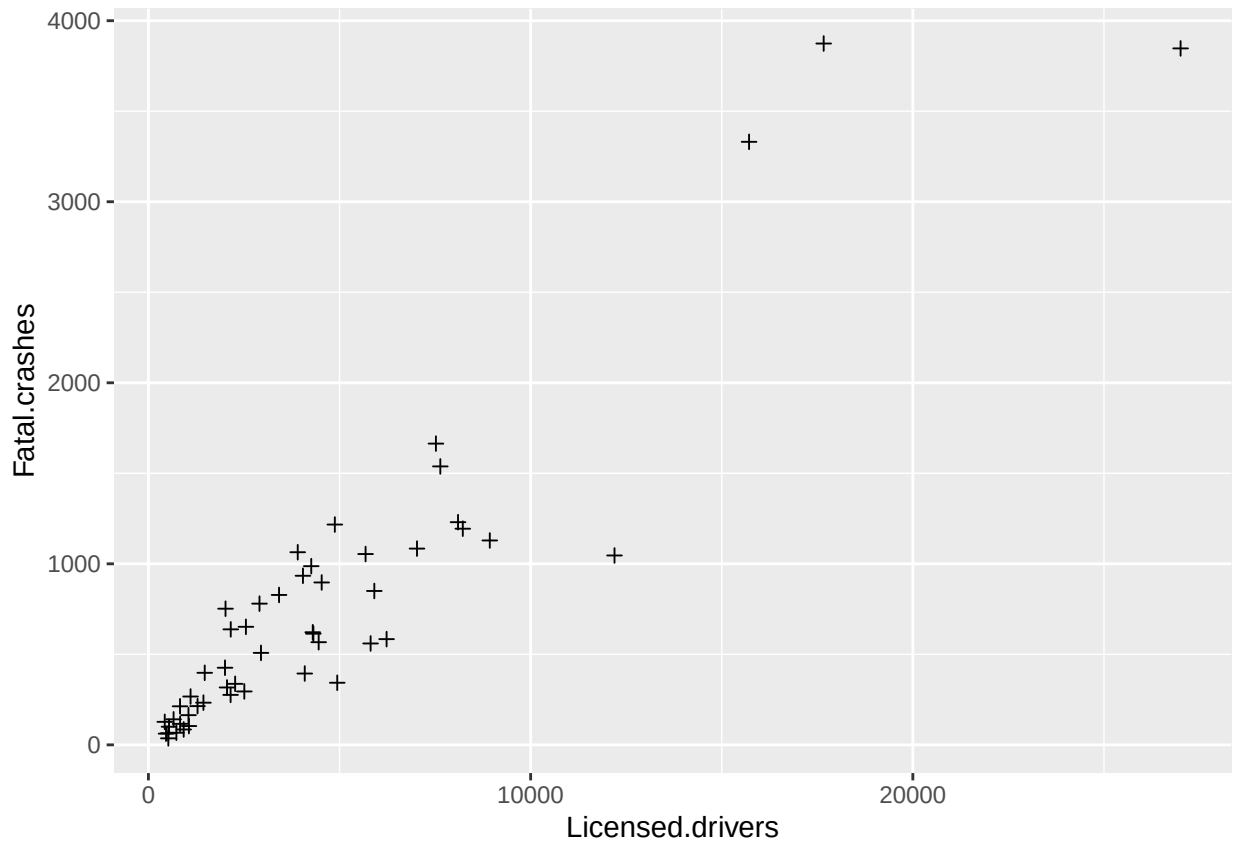
Complete the following problems and submit them to Gradescope. Your work will be manually graded for this assignment, so there are no strict code formatting requirements.

Problem 1

The data file `state_crashes.csv` contains the number of fatal accidents and licensed drivers (in thousands) in 2020 in each state. Additional information regarding whether each state has passed laws banning hand-held electronic devices while driving - no (1), for some drivers (2), yes (3) - is given.

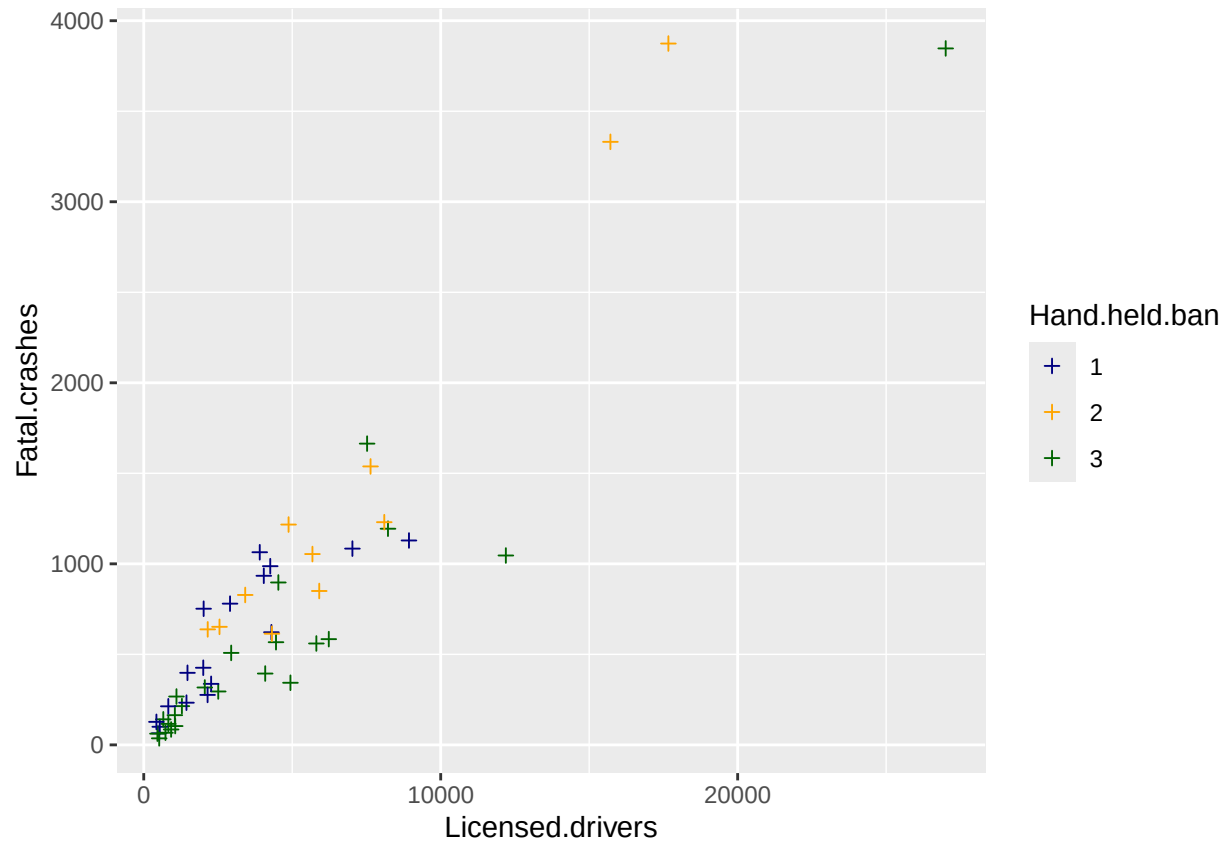
a

Plot the licensed drivers (horizontal axis) and fatal accidents (vertical axis) data in a scatterplot using an addition sign (+) as the points.



b

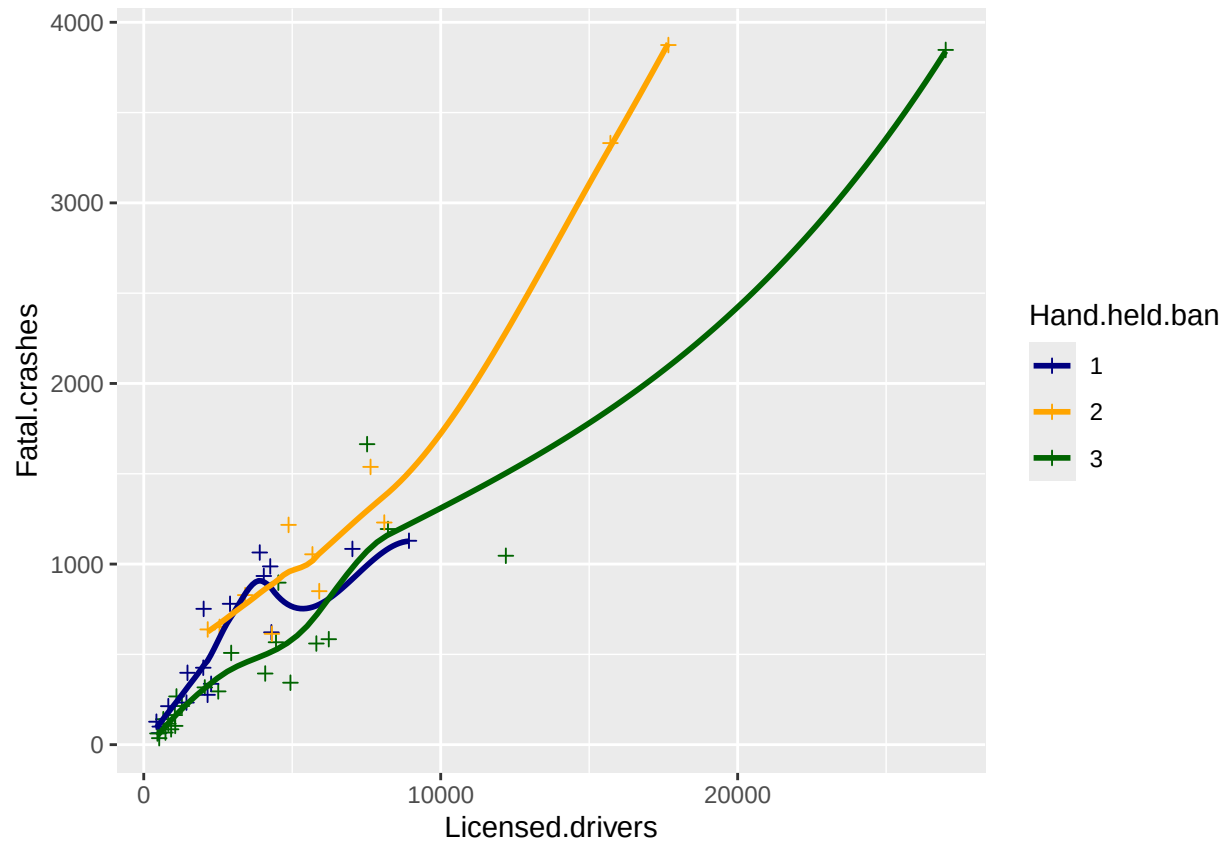
Modify the graphic created in part a to distinguish the states that have no, partial, and full hand-held electronic bans with navy blue, orange, and dark green, respectively.



c

Add a smoothing line for each group of states to the graphic created in part b (no confidence bands).

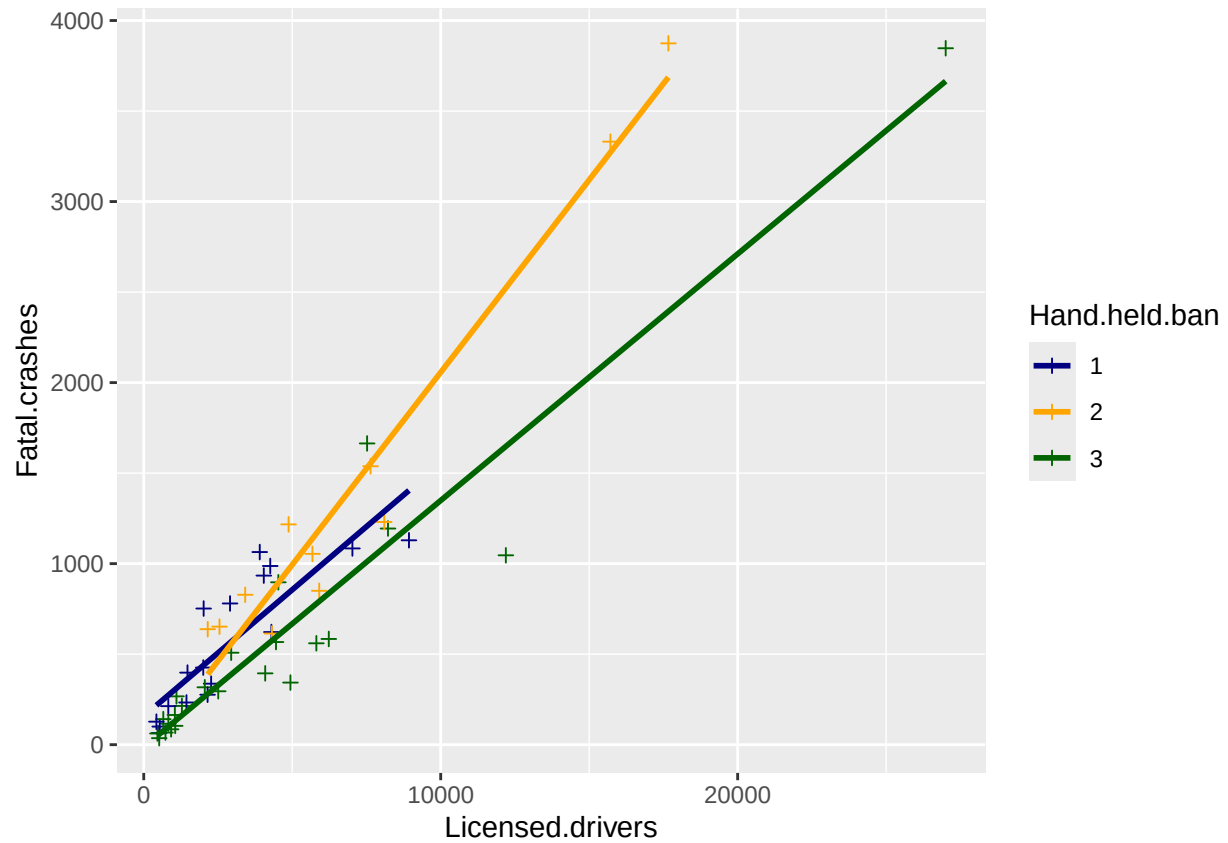
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



d

Modify the smoothing lines for each group of states in the graphic created in part c to regression lines.

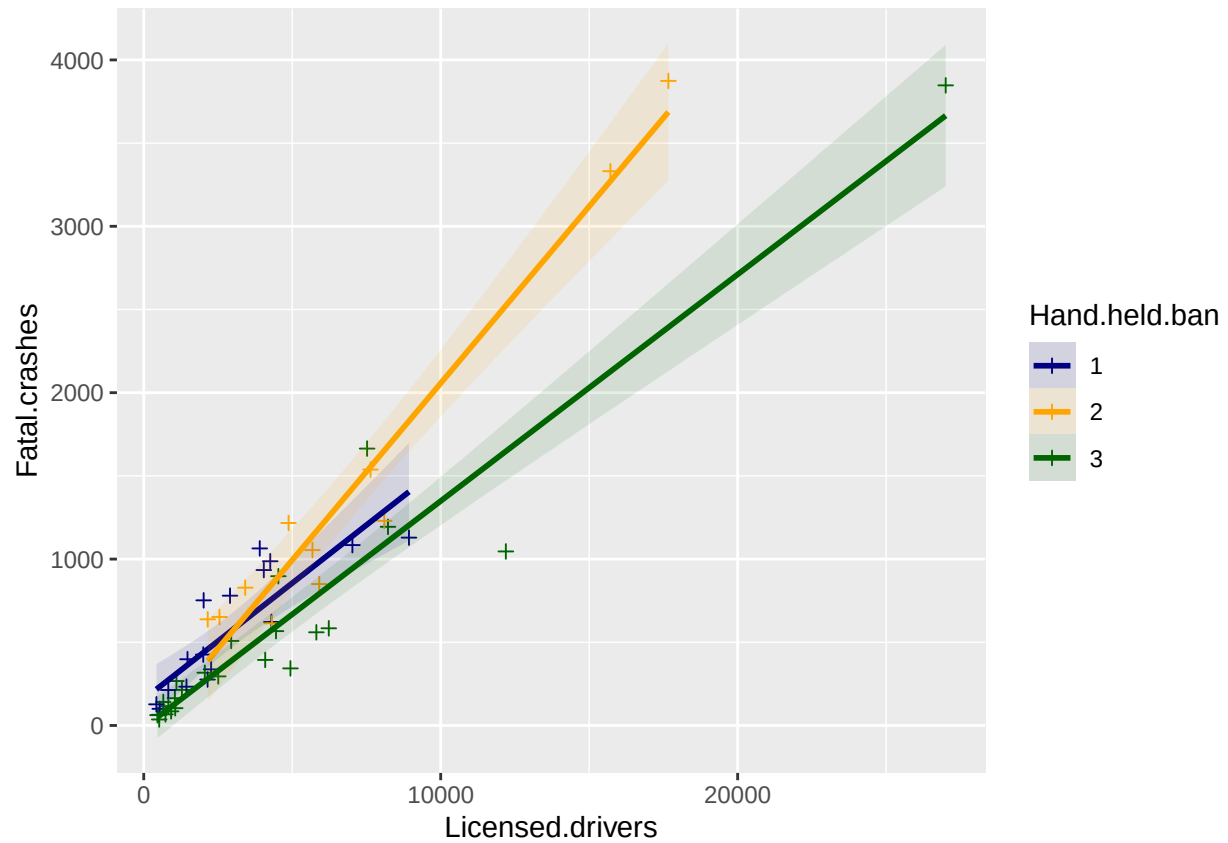
```
## 'geom_smooth()' using formula = 'y ~ x'
```



e

Add confidence bands to the graphic created in part d that are shaded the same color as the line and are mostly transparent.

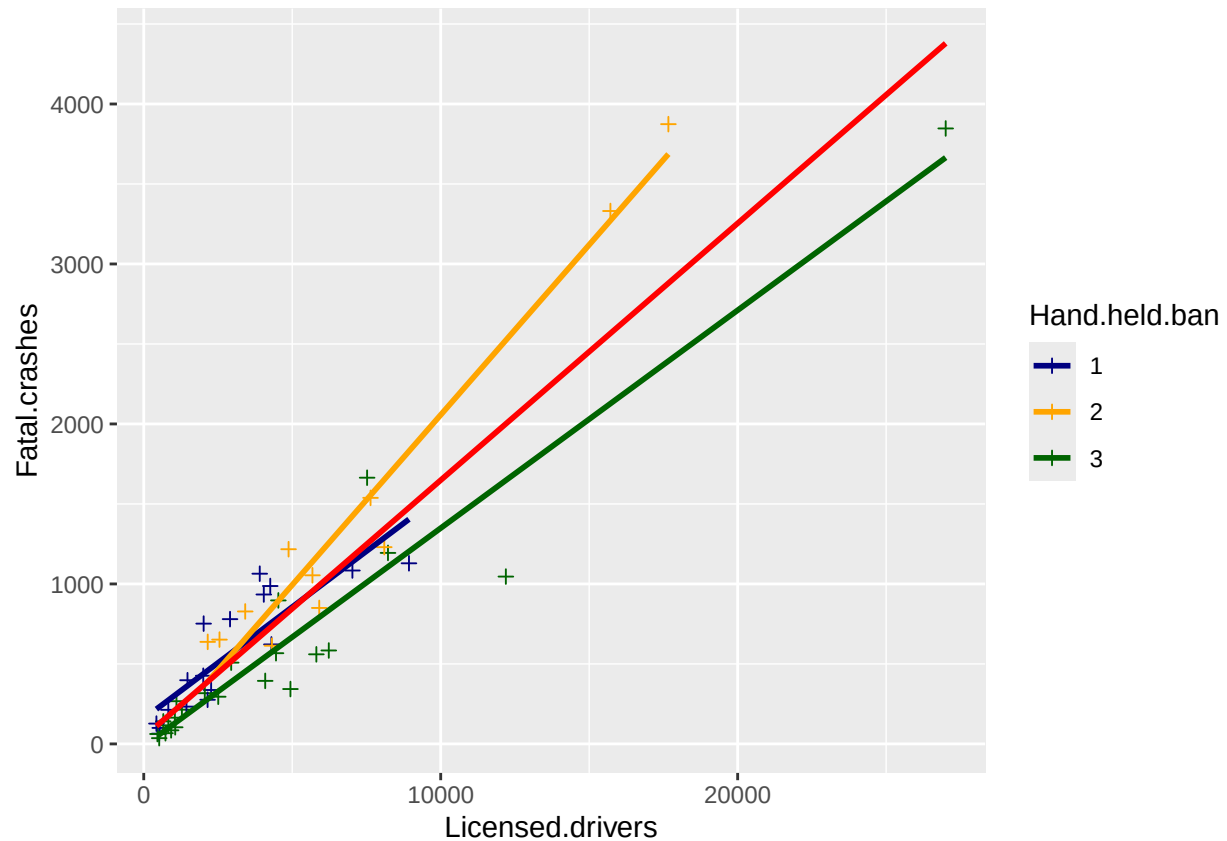
```
## 'geom_smooth()' using formula = 'y ~ x'
```



f

Add a single red regression line that describes all states to the graphic created in part d.

```
## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
```



g

What conclusions can you draw from this set of graphics?

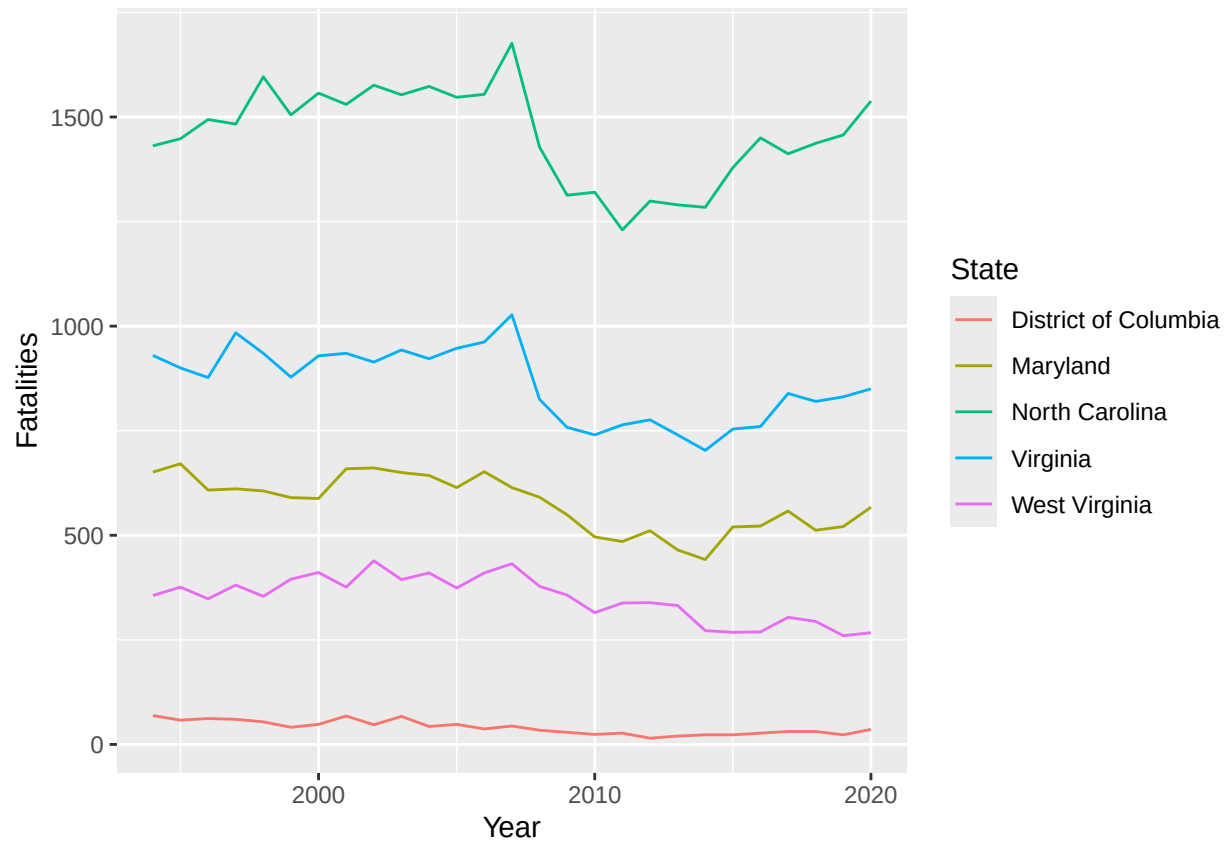
Note: You should not use the word *cause*.

Problem 2

The data file `fatalities.csv` contains the number of fatalities from car crashes from 1994 through 2020 in Virginia and surrounding states. This file also contains the resident population total (in thousands), the number of licensed drivers (in thousands), the number of registered vehicles (in thousands), and the number of vehicle miles driven (in billions).

a

Plot the number of fatalities each year distinguishing each state by line color.

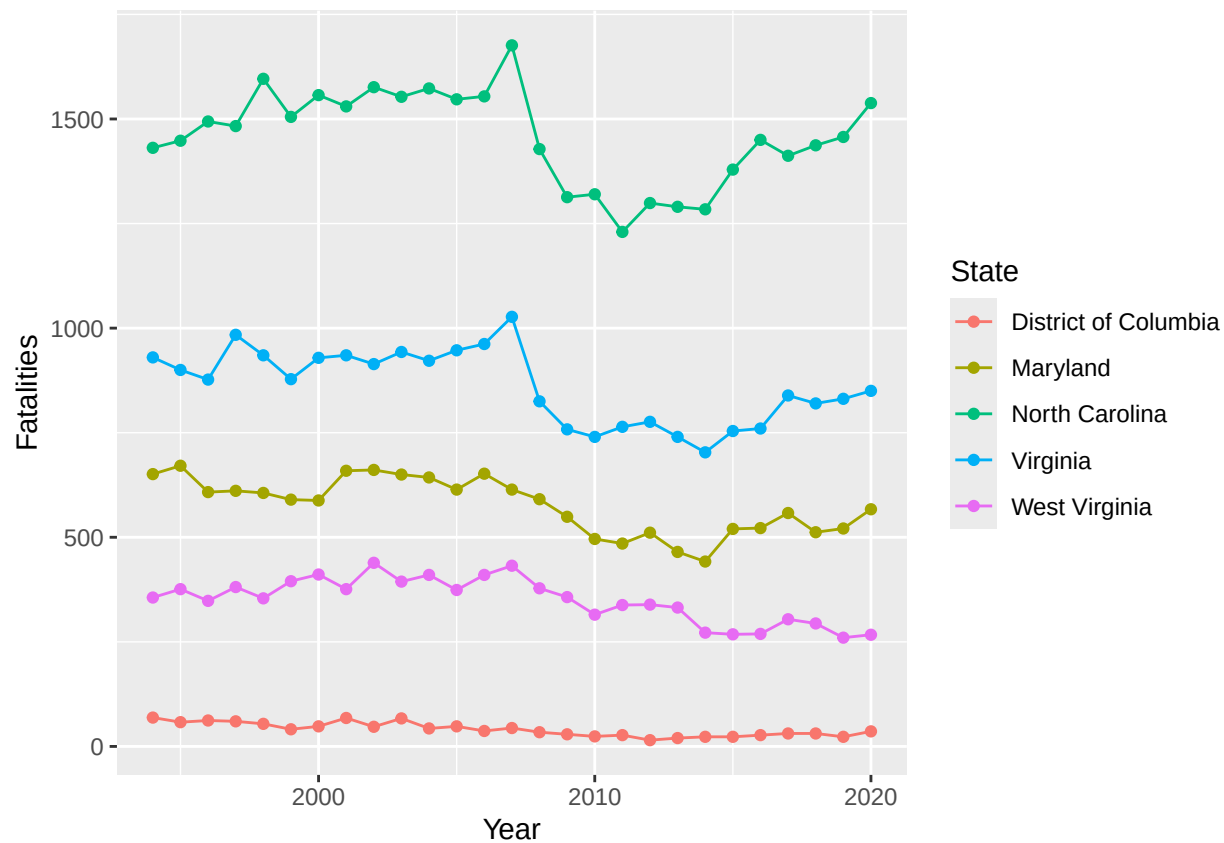


b

What conclusions might you draw from the plot created in part a? Do the states seem to have similar trends over time?

c

Add points that are the same colors as the lines to the graphic created in part a.



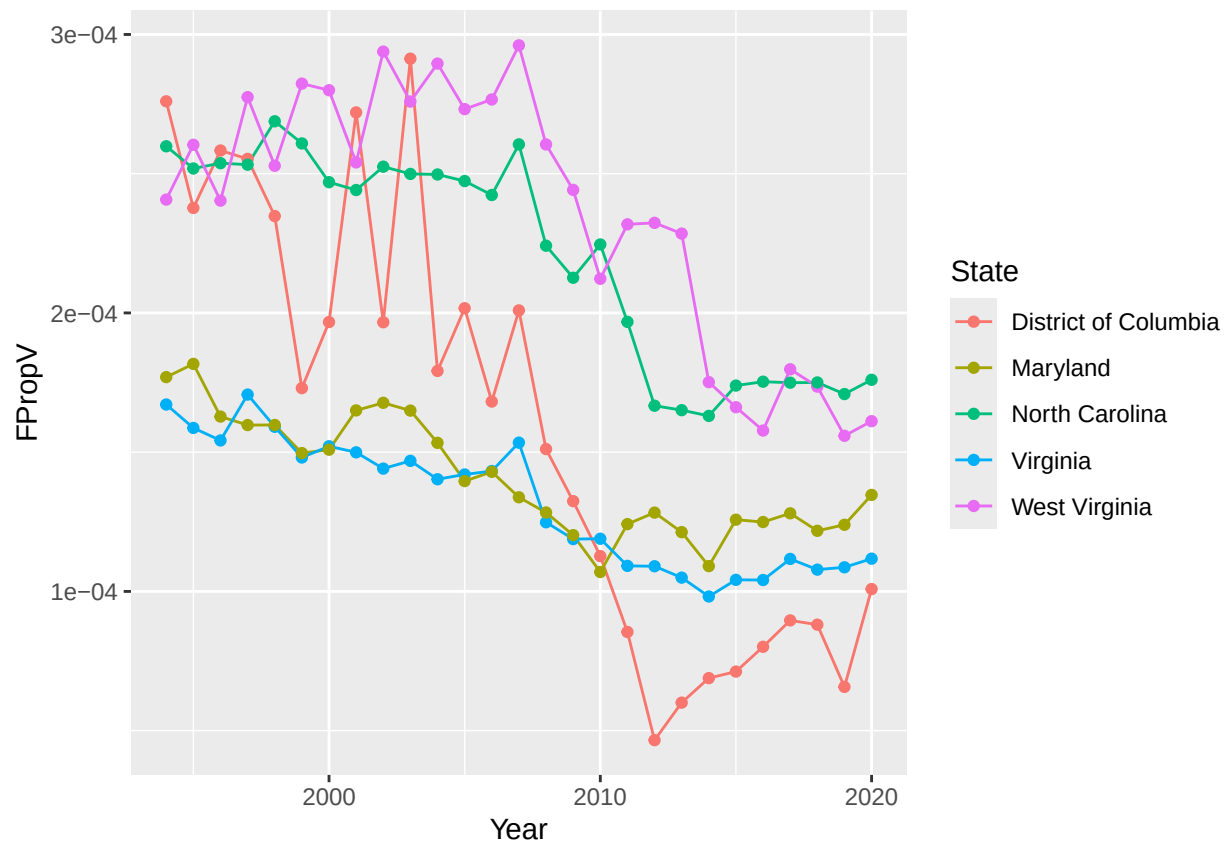
d

Explain one advantage to using the plot created in part c instead of the plot created in part a.

e

Modify the plot in part c to show the proportion of fatalities with respect to the total number of vehicles registered.

Note: You will need to account for the scaling of the data to show the accurate proportions.



f

Does this plot change any of your conclusions drawn in part b? Explain.

g

Modify the plot in part c to show the proportion of fatalities considering the total vehicle miles driven.

Note: You will need to account for the scaling of the data to show the accurate proportions.

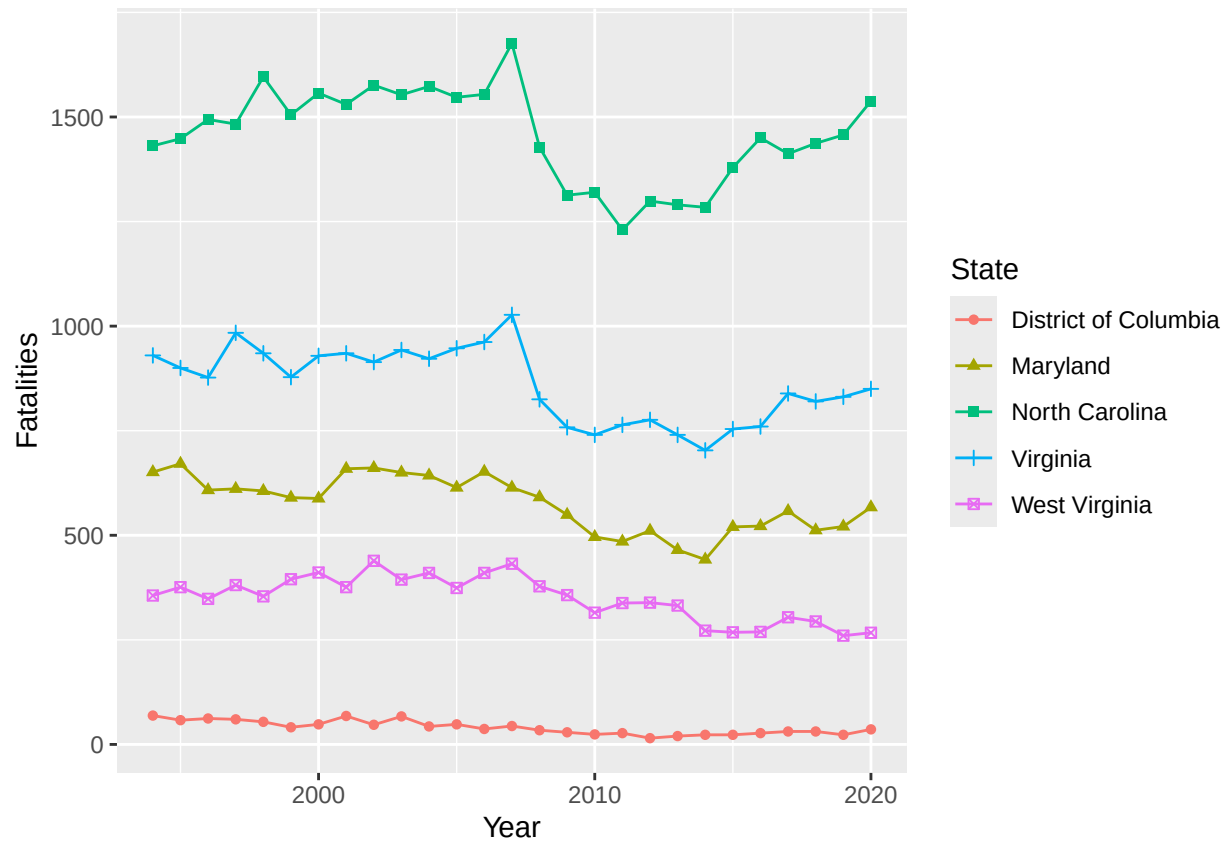


h

Describe any differences you see between the plots produced in parts e and g. Provide potential reasons for these differences.

i

Modify the point type in the graphic created in part c to vary by state.



j

Does varying the point type by state provide significant benefit in communicating information?